

Project 1 — Diabetes Diagnosis Prediction

Professor Submission Report

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1. Project Overview

This project develops and evaluates a binary classification model for predicting the diabetes outcome contained in the provided dataset. The work emphasizes reproducible machine-learning practice, transparent model evaluation, explainability, threshold analysis, artifact integrity, and responsible-AI limitations.

2. Dataset

Property	Value
Rows	768
Columns	9
Predictive features	8
Target	Outcome
Outcome = 0	500
Outcome = 1	268
Dataset SHA256	698c203a14aa31941d2251175330c9199f3ccdb31597abbba2a3e35416257a72

3. Data Quality and Clinical EDA

The dataset contains no pandas-detected missing values and no duplicate rows. Several physiological variables contain zero values that are clinically suspicious and were therefore examined separately from legitimate zero-valued pregnancy counts. Suspicious zeros were identified in Glucose, BloodPressure, SkinThickness, Insulin, and BMI.

The exploratory analysis showed that Glucose had the strongest positive correlation with the Outcome target, followed by BMI, Age, Pregnancies, and DiabetesPedigreeFunction. These relationships are statistical associations within this dataset and are not causal conclusions.

4. Train/Test Strategy

Property	Value
Training observations	614
Test observations	154
Test proportion	20%
Random state	42
Stratification	Yes
Predictive features	['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age']

The test set was held out from model fitting. Model preprocessing and fitting were performed using the training data only.

5. Model Development

Two pipelines were evaluated. The baseline pipeline used StandardScaler followed by LogisticRegression. The clinical preprocessing pipeline additionally converted clinically suspicious zero values to missing values and applied median imputation before scaling and logistic regression.

The baseline Logistic Regression model achieved the strongest overall holdout performance and was selected as the champion using ROC-AUC as the primary selection metric, followed by recall, F1, and accuracy as secondary considerations.

6. Champion Model Evaluation

Metric	Champion Score
Accuracy	0.7143
Precision	0.6087
Recall	0.5185
F1	0.5600
ROC-AUC	0.8230

The model achieved a holdout ROC-AUC of approximately 0.8230. The test set is relatively small, so these results should be interpreted as an academic evaluation rather than evidence of clinical effectiveness.

7. Model Explainability

The champion model is a standardized Logistic Regression model. Its coefficients provide an interpretable linear relationship between standardized feature values and the model's log-odds output.

The strongest positive coefficient was associated with Glucose, followed by BMI and Pregnancies. Coefficients represent statistical associations in the fitted model and should not be interpreted as causal effects or medical rules.

8. Threshold Analysis

A probability threshold grid from 0.20 through 0.80 was evaluated. Within the evaluated grid, a threshold of 0.25 produced the highest observed recall and F1 on the held-out test set. However, no clinical operating threshold was selected.

Clinical threshold selection requires intended-use definition, error-cost analysis, calibration, appropriate clinical validation, and consideration of the consequences of false-positive and false-negative predictions.

9. Final Model Artifact

Property	Validated Value
Artifact	diabetes_baseline_logistic_regression.joblib
Size	1,953 bytes
SHA256	5db1b088d1bfb18c74c9e7622e013c4258759f9f6f9e34a6a4d4398f8febba9c
Dataset SHA256	698c203a14aa31941d2251175330c9199f3ccdb31597abbba2a3e35416257a72
Integrity	PASS

The final artifact was independently loaded and validated. Its SHA256 hash matches the Step 9 evidence, Step 10 expected hash, Step 10 actual hash, and model metadata.

10. Reproducibility Validation

Independent validation confirmed that the saved model loads successfully, contains the expected eight features, produces predictions for all 154 held-out observations, produces valid probabilities, and reproduces the recorded evaluation metrics. The dataset and model hashes also match their recorded evidence.

11. Responsible AI and Limitations

This project is an academic and portfolio machine-learning exercise. It is not a clinically validated diagnostic system and must not be used to make medical decisions.

Important limitations include the small dataset size, the limited holdout test set, suspicious zero values in several physiological variables, lack of external validation, lack of prospective clinical validation, and absence of clinical calibration and utility analysis.

The project therefore demonstrates machine-learning methodology, reproducibility, interpretability, and responsible-AI thinking rather than clinical readiness.

12. Final Conclusion

The project successfully produced a reproducible Logistic Regression classification artifact and a complete academic evaluation package. The final model, dataset, evidence, and notebook passed integrity and reproducibility checks.

Final Project Status: PASS